Web Appendix for "A Bayesian Critique of a Rank-Based Method for Surrogate Marker Evaluation"

A. Derivation of θ and δ as Expectations over the Covariate Distribution

This appendix derives expressions for θ and δ as expectations over the covariate distribution.

A.1 *Derivation of θ as an Expectation over the Covariate Distribution*

Under the model in Equation (3), it is easy to see that

$$Y_{1i} - Y_{0i} \mid X_i = x \overset{\text{ind.}}{\sim} \mathcal{N}\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right), \quad i = 1, \ldots, n, \ \forall\, x \in \mathcal{X}.$$

Therefore, we obtain this analytical expression for V^x_Y:

$$\begin{aligned}
V^x_Y &= \mathbb{P}\left(Y_{1i} > Y_{0i} \mid X_i = x\right) \\
&= \mathbb{P}\left(Y_{1i} - Y_{0i} > 0 \mid X_i = x\right) \\
&= \mathbb{P}\left(\frac{Y_{1i} - Y_{0i} - \mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}} > \frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}} \mid X_i = x\right) \\
&= \mathbb{P}\left(W > \frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) \\
&= 1 - \Phi\left(\frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) \\
&= \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right), \ \forall\, x \in \mathcal{X},
\end{aligned}$$

where $W \sim \mathcal{N}(0,1)$ and Φ is the standard normal CDF. Then, by the law of total probability, we can derive an analytical expression for V_Y as follows:

$$\begin{aligned}
V_Y &= \mathbb{P}\left(Y_{1i} > Y_{0i}\right) \\
&= \int_{\mathcal{X}} V^x_Y \, f(x) \, \mathrm{d}x \\
&= \int_{\mathcal{X}} \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) f(x) \, \mathrm{d}x.
\end{aligned}$$

Applying a similar logic to V_S, we obtain this analytical expression for θ:

$$\theta = \int_{\mathcal{X}} \theta^x \, f(x) \, \mathrm{d}x,$$

where we denote

$$\theta^x = \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) - \Phi\left(\frac{\mu^x_{S_1} - \mu^x_{S_0}}{\sqrt{\Sigma^x_{S_1S_1} + \Sigma^x_{S_0S_0} - 2\Sigma^x_{S_1S_0}}}\right), \ \forall\, x \in \mathcal{X}.$$

A.2 *Derivation of δ as an Expectation over the Covariate Distribution*

Under the model in Equation (3), it is easy to see that

$$Y_{1i} - Y_{0j} \mid X_i = x_i, X_j = x_j \sim \mathcal{N}\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right), \ \forall\, x_i, x_j \in \mathcal{X},\ i, j = 1, \ldots, n.$$

Therefore, we obtain this analytical expression for $U_Y^{x_ix_j}$:

$$\begin{aligned}
U_Y^{x_ix_j} &= \mathbb{P}\left(Y_{1i} > Y_{0j} \mid X_i = x_i, X_j = x_j\right) \\
&= \mathbb{P}\left(\frac{Y_{1i} - Y_{0j} - \mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}} > \frac{-\mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}} \mid X_i = x_i, X_j = x_j\right) \\
&= \mathbb{P}\left(W > \frac{-\mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) \\
&= 1 - \Phi\left(\frac{-\mu_{Y_1}^{x_i} + \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) \\
&= \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right), \quad \forall\, x_i, x_j \in \mathcal{X},\ i, j = 1, \ldots, n.
\end{aligned}$$

Then, by the law of total probability, we can derive an analytical expression for U_Y as follows:

$$\begin{aligned}
U_Y &= \mathbb{P}\left(Y_{1i} > Y_{0j}\right) \\
&= \int_{\mathcal{X}}\int_{\mathcal{X}} U_Y^{x_i,x_j}\, f(x_i)\, f(x_j)\, \mathrm{d}x_i\, \mathrm{d}x_j \\
&= \int_{\mathcal{X}}\int_{\mathcal{X}} \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) f(x_i)\, f(x_j)\, \mathrm{d}x_i\, \mathrm{d}x_j.
\end{aligned}$$

Applying a similar logic to U_S, we obtain this analytical expression for δ:

$$\delta = \int_{\mathcal{X}}\int_{\mathcal{X}} \delta^{x_ix_j}\, f(x_i)\, f(x_j)\, \mathrm{d}x_i\, \mathrm{d}x_j,$$

where we denote

$$\delta^{x_ix_j} = \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}}\right) - \Phi\left(\frac{\mu_{S_1}^{x_i} - \mu_{S_0}^{x_j}}{\sqrt{\Sigma_{S_1S_1}^{x_i} + \Sigma_{S_0S_0}^{x_j}}}\right), \ \forall\, x_i, x_j \in \mathcal{X}.$$

A.3 *Derivation of θ and δ in the Condionally Orthogonal and Homoscedastic Setting*

Under the assumption of conditional orthogonality stated in the paper, it follows that

$$V_Y^x = U_Y^x \text{ and } V_S^x = U_S^x,\ \forall\, x \in \mathcal{X},$$

so that

$$\theta^x = \delta^{xx}\ \forall\, x \in \mathcal{X}. \tag{A.1}$$

Under the assumption of conditional homoscedasticity stated in the paper, it follows that

$$\theta = \mathbb{E}_{X_i \sim f}(g(X_i, X_i))\,, \quad \delta = \mathbb{E}_{(X_i, X_j) \sim f^2}(g(X_i, X_j))\,,$$

where we denote

$$g(x_i, x_j) = \Phi\left(\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}\right) - \Phi\left(\mu_{S_1}^{x_i} - \mu_{S_0}^{x_j}\right)\ \forall\, x_i, x_j \in \mathcal{X}. \tag{A.2}$$

B. Derivation of $|\theta - \delta|$ in the Binary Covariate Setting

Under the setting described in Section 2.4.1 of the main paper, θ and δ reduce to

$$\theta = \frac{\theta^0 + \theta^1}{2}, \quad \delta = \frac{\delta^{00} + \delta^{11} + \delta^{01} + \delta^{10}}{4}.$$

Based on Equation (A.1), we have that

$$|\theta - \delta| = \frac{|\delta^{00} + \delta^{11} - \delta^{01} - \delta^{10}|}{4}. \tag{A.3}$$

This expression highlights that the distance between the two parameters is driven by differences between comparisons within the same covariate group and comparisons across different covariate groups. Importantly, this effect does not arise from treatment effect heterogeneity.

To derive a formula for $|\theta - \delta|$, let us further simplify the setting by assuming that S is a perfect surrogate for Y with homogeneous treatment effects across covariate groups:

$$\mu_{Y_1}^x - \mu_{Y_0}^x = \mu_{S_1}^x - \mu_{S_0}^x = \Delta > 0 \quad \forall\, x \in \{0, 1\}.$$

In this setting, we can easily compute the terms in Equation (A.3) using Equation (A.2):

$$\delta^{00} = \Phi(\Delta) - \Phi(\Delta) = 0,$$

$$\delta^{11} = \Phi(\Delta) - \Phi(\Delta) = 0,$$

$$\delta^{01} = \Phi\big(\Delta + \mu^0_{Y_0} - \mu^1_{Y_0}\big) - \Phi\big(\Delta + \mu^0_{S_0} - \mu^1_{S_0}\big)\,,$$

$$\delta^{10} = \Phi\big(\Delta + \mu^1_{Y_0} - \mu^0_{Y_0}\big) - \Phi\big(\Delta + \mu^1_{S_0} - \mu^0_{S_0}\big)\,.$$

Hence, we conclude that

$$|\theta - \delta| = \frac{1}{4}\left|\Phi(\Delta + d_Y) - \Phi(\Delta + d_S) + \Phi(\Delta - d_Y) - \Phi(\Delta - d_S)\right|,$$

where we denote $d_Y = \mu^0_{Y_0} - \mu^1_{Y_0}$ and $d_S = \mu^0_{S_0} - \mu^1_{S_0}$.

B.1 *Supremum of* $|\theta - \delta|$

We now prove analytically that $\sup_{d_Y, d_S}\{|\theta - \delta|\} = \frac{1}{4}|1 - 2\Phi(\Delta)|$.

DO THE MATH!

C. Derivation of θ and δ in the Gaussian Covariate Setting

Using the setting from Section 2.4.2, we express θ and δ via Equation (A.2) as follows:

$$\theta = \mathbb{E}_{X_i \sim f}(\Phi\left(X_i(\beta_{Y_1} - \beta_{Y_0})\right) - \Phi\left(X_i(\beta_{S_1} - \beta_{S_0})\right))\,,$$

$$\delta = \mathbb{E}_{(X_i, X_j) \sim f^2}(\Phi\left(X_i\beta_{Y_1} - X_j\beta_{Y_0}\right) - \Phi\left(X_i\beta_{S_1} - X_j\beta_{S_0}\right))\,.$$

C.1 *Derivation of* θ

Using the perfect surrogacy assumption in Equation (4), we have that

$$\begin{aligned}\theta &= \mathbb{E}_{X_i \sim f}(\Phi\left(X_i(\beta_{Y_1} - \beta_{Y_0})\right) - \Phi\left(X_i(\beta_{S_1} - \beta_{S_0})\right)) \\ &= \mathbb{E}_{X_i \sim f}(\Phi\left(X_i\Delta\right) - \Phi\left(X_i\Delta\right)) = 0.\end{aligned}$$

C.2 *Derivation of* δ

Observe that by the Gaussian covariate assumption, we have that

$$X_i\beta_{Y_1} - X_j\beta_{Y_0} \sim \mathcal{N}\left(m(\beta_{Y_1} - \beta_{Y_0}), s^2(\beta^2_{Y_1} + \beta^2_{Y_0})\right), \quad i, j = 1, \ldots, n,$$

$$X_i\beta_{S_1} - X_j\beta_{S_0} \sim \mathcal{N}\left(m(\beta_{S_1} - \beta_{S_0}), s^2(\beta^2_{S_1} + \beta^2_{S_0})\right), \quad i, j = 1, \ldots, n.$$

Therefore, we can derive an expanded analytical expression for δ as follows:

$$\begin{aligned}\delta &= \mathbb{E}_{(X_i,X_j)\sim f^2}(\Phi\left(X_i\beta_{Y_1} - X_j\beta_{Y_0}\right)) - \mathbb{E}_{(X_i,X_j)\sim f^2}(\Phi\left(X_i\beta_{S_1} - X_j\beta_{S_0}\right)) \\ &= \Phi\left(\frac{m(\beta_{Y_1} - \beta_{Y_0})}{\sqrt{1 + s^2(\beta_{Y_1}^2 + \beta_{Y_0}^2)}}\right) - \Phi\left(\frac{m(\beta_{S_1} - \beta_{S_0})}{\sqrt{1 + s^2(\beta_{S_1}^2 + \beta_{S_0}^2)}}\right),\end{aligned}$$

where the last line follows from the following property of Gaussian random variables.

LEMMA 1: *Let $W \sim \mathcal{N}(m, s^2)$ for some $m \in \mathbb{R}$ and $s \in \mathbb{R}_+$. Then, we have that*

$$\mathbb{E}_{W\sim\mathcal{N}(m,s^2)}(\Phi(W)) = \Phi\left(\frac{m}{\sqrt{1+s^2}}\right).$$

C.3 *Supremum of $|\theta - \delta|$*

DO THE MATH!

D. Regularity and Posterior Consistency

This appendix provides a formal definition of regularity for Bayesian models and states Schwartz's theorem, which is a fundamental result on posterior consistency of regular models.

DEFINITION 1 (Regularity of a Bayesian model): Let a Bayesian model be defined as

$$\begin{aligned} D_i \mid \phi &\overset{\text{i.i.d.}}{\sim} f_\phi, \quad i = 1, \ldots, n, \\ \phi &\sim \pi, \end{aligned}$$

where D_i is the observed data for unit i, f_ϕ is the likelihood function parametrized by ϕ, and π is a prior distribution over the parameter ϕ. We say that the Bayesian model is *regular* if the following conditions hold:

- **Kullback-Leibler support of the prior**: The prior π assigns positive mass to every Kullback-Leibler neighborhood of the true parameter ϕ^*:

  $$\pi\left(\{\phi : D_{\mathrm{KL}}(f_{\phi^*} \| f_\phi) < \varepsilon\}\right) > 0 \quad \forall\, \varepsilon \in \mathbb{R}_+,$$

  where $D_{\mathrm{KL}}(f_{\phi^*} \| f_\phi)$ is the Kullback-Leibler divergence of f_ϕ from f_{ϕ^*}.

- **Identifiability**: The model is identifiable:

$$f_\phi \neq f_{\phi'} \quad \forall\, \phi \neq \phi'.$$

THEOREM 1 (Schwartz's theorem (Schwartz, 1965)): *Let the data be generated from the following process:*

$$D_i \overset{i.i.d.}{\sim} f_{\phi^*}, \quad i = 1, \ldots, n,$$

and let the Bayesian model be defined as

$$D_i \mid \phi \overset{i.i.d.}{\sim} f_\phi, \quad i = 1, \ldots, n,$$

$$\phi \sim \pi.$$

If the model is regular according to Definition 1, then we have that

$$\pi\left(U^c \mid D_{1:n}\right) \overset{P_{\phi^*}}{\to} 0,$$

for every neighborhood U of ϕ^*, where P_{ϕ^*} is the probability measure induced by f_{ϕ^*}.

E. Derivation of the Bayes Factor for the Validation Threshold

Under the model described in Section 3.3 of the main paper, the marginal likelihoods under the null and alternative hypotheses can be explicitly stated as follows:

$$\begin{aligned}
L\Big(\widehat{V}_S \mid H_0\Big) &= \binom{n}{n\,\widehat{V}_S} \left(\frac{1}{2}\right)^{n\,\widehat{V}_S} \left(1 - \frac{1}{2}\right)^{n - n\,\widehat{V}_S} \\
&= \binom{n}{n\,\widehat{V}_S} \frac{1}{2^n}, \\
L\Big(\widehat{V}_S \mid H_1\Big) &= \int_{\frac{1}{2}}^{1} \binom{n}{n\,\widehat{V}_S} v_S^{n\,\widehat{V}_S} \left(1 - v_S\right)^{n - n\,\widehat{V}_S} \frac{v_S^{a-1}\left(1 - v_S\right)^{b-1}}{B(a,b)\left(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\right)} \,\mathrm{d}v_S \\
&= \binom{n}{n\,\widehat{V}_S} \frac{B\Big(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\Big)\left(1 - F_{\mathrm{Beta}\left(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\right)}\left(\frac{1}{2}\right)\right)}{B(a,b)\left(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\right)},
\end{aligned}$$

where $L\Big(\widehat{V}_S \mid H_0\Big)$ and $L\Big(\widehat{V}_S \mid H_1\Big)$ are the marginal likelihoods under the null and alternative hypotheses, respectively, $B(a,b)$ is the Beta function and $F_{\mathrm{Beta}(a,b)}$ is the CDF of the

Beta(a, b) distribution. Then, the Bayes factor in favor of the alternative hypothesis is

$$
\begin{aligned}
BF_n &= \frac{L\Big(\widehat{V}_S \mid H_1\Big)}{L\Big(\widehat{V}_S \mid H_0\Big)} \\
&= \frac{B\Big(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\Big)\,\Big(1 - F_{\mathrm{Beta}\left(a+n\,\widehat{V}_S,\, b+n-n\,\widehat{V}_S\right)}\left(\frac{1}{2}\right)\Big)}{B(a,b)\,\Big(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\Big)}\, 2^n.
\end{aligned}
$$

Notice that the distribution of BF_n under H_0 has closed form: since $n\,\widehat{V}_S \in \{0, \ldots, n\}$, the Bayes factor takes at most $n+1$ distinct values $x_0, \ldots, x_n$ with

$$
\mathbb{P}(BF_n = x_k \mid H_0) = \binom{n}{k}\frac{1}{2^n}, \quad k = 0, \ldots, n,
$$

where

$$
x_k = \frac{B(a+k,\, b+n-k)\,\Big(1 - F_{\mathrm{Beta}(a+k,\, b+n-k)}\left(\frac{1}{2}\right)\Big)}{B(a,b)\,\Big(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\Big)}\, 2^n, \quad k = 0, \ldots, n.
$$

Finally, to compute the probability on the left-hand side of Equation (11), we observe that

$$
\mathbb{P}\left(BF_n = x_k \mid V_S = v_S > \frac{1}{2}\right) = \binom{n}{k} v_S^k (1 - v_S)^{n-k}, \quad k = 0, \ldots, n.
$$

F. Simulation Study: Data-Generating Processes and Implementation Details

This appendix describes the data-generating processes and the Markov chain Monte Carlo implementation details for the simulation study in Section 4 of the main paper.

F.1 *Data-Generating Processes*

We list the data-generating processes for the six settings considered in the simulation study.

- **Setting 1 (Useless surrogate):** The potential outcomes are generated as follows:

$$
\begin{aligned}
Y_{1i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(5, 3), \quad S_{1i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(\tfrac{2}{3}, 1\right), \\
Y_{0i} &\overset{\text{i.i.d.}}{\sim} \mathcal{N}(3, 3), \quad S_{0i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(\tfrac{2}{3}, 1\right),
\end{aligned}
\qquad i = 1, \ldots, n.
$$

  Therefore, by the properties of Gaussian distributions, we have that

$$
P_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu^*, \Sigma^*), \quad i = 1, \ldots, n,
$$

where

$$\mu^* = \begin{bmatrix} 5 \\ \frac{2}{3} \\ 3 \\ \frac{2}{3} \end{bmatrix}, \qquad \Sigma^* = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Setting 2 (Perfect surrogate):** The potential outcomes are generated as follows:

$$\begin{aligned} &Y_{1i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(6,3), \quad S_{1i} \mid Y_{1i} \overset{\text{ind.}}{\sim} \mathcal{N}(Y_{1i}, 0.1), \\ &Y_{0i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(2.5,3), \quad S_{0i} \mid Y_{0i} \overset{\text{ind.}}{\sim} \mathcal{N}(Y_{0i}, 0.1), \end{aligned} \qquad i = 1, \ldots, n.$$

Therefore, by the properties of Gaussian distributions, we have that

$$P_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu^*, \Sigma^*), \quad i = 1, \ldots, n,$$

where

$$\mu^* = \begin{bmatrix} 6 \\ 6 \\ 2.5 \\ 2.5 \end{bmatrix}, \qquad \Sigma^* = \begin{bmatrix} 3 & 3 & 0 & 0 \\ 3 & 3.1 & 0 & 0 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3.1 \end{bmatrix}.$$

- **Setting 3 (Imperfect surrogate):** The potential outcomes are generated as follows:

$$\begin{aligned} &S_{1i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(5,3), \quad Y_{1i} \mid S_{1i} \overset{\text{ind.}}{\sim} \mathcal{N}(S_{1i} + 1.5,\, 0.6), \\ &S_{0i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(3,3), \quad Y_{0i} \mid S_{0i} \overset{\text{ind.}}{\sim} \mathcal{N}(S_{0i},\, 0.6), \end{aligned} \qquad i = 1, \ldots, n.$$

Therefore, by the properties of Gaussian distributions, we have that

$$P_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu^*, \Sigma^*), \quad i = 1, \ldots, n,$$

where

$$\mu^* = \begin{bmatrix} 6.5 \\ 5 \\ 3 \\ 3 \end{bmatrix}, \qquad \Sigma^* = \begin{bmatrix} 3.6 & 3 & 0 & 0 \\ 3 & 3 & 0 & 0 \\ 0 & 0 & 3.6 & 3 \\ 0 & 0 & 3 & 3 \end{bmatrix}.$$

Setting 4 (Misspecified model).. The surrogate potential outcomes are log-normal: $S_{1i} \sim \exp(\mathcal{N}(2.5,\, 1.5^2))$ and $S_{0i} \sim \exp(\mathcal{N}(0.5,\, 1.5^2))$. The primary outcome is a non-linear function of the surrogate plus log-normal noise:

$$Y_{1i} = 2.0 + 1.2\sqrt{S_{1i}} + 0.3\exp(S_{1i}/500) + \exp(\varepsilon_{1i}),$$
$$Y_{0i} = 0.8\sqrt{S_{0i}} + 0.2\exp(S_{0i}/50) + \exp(\varepsilon_{0i}),$$

where $\varepsilon_{gi} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\, 0.3)$. This setting is designed to violate the Gaussian parametric assumptions of our Bayesian model.

Setting 5.. A binary baseline covariate $X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5)$ is generated, and the distribution of $P_i \mid X_i$ is the following mixture:

$$P_i \mid X_i \overset{\text{ind.}}{\sim} X_i\, \mathcal{N}_4\big(\mu^1, \Sigma^1\big) + (1 - X_i)\, \mathcal{N}_4\big(\mu^0, \Sigma^0\big)\,, \quad i = 1, \ldots, n,$$

with

$$\mu^0 = (5, 5, 0, 0)^\top, \qquad \mu^1 = (5, -5, 0, -10)^\top,$$
$$\Sigma^0 = \Sigma^1 = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}, \qquad A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Setting 6.. A Gaussian baseline covariate $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(3, 1)$ is generated, and the distribution of $P_i \mid X_i$ follows a linear Gaussian model:

$$P_i \mid X_i \overset{\text{ind.}}{\sim} \mathcal{N}_4(X_i\, \beta,\, \Sigma)\,, \quad i = 1, \ldots, n,$$

with

$$\beta = (1,\, 7,\, 0,\, 6)^\top, \qquad \Sigma = \tfrac{1}{2}\, I_4.$$

Implementation details.. For the Bayesian method, we ran $T = 500$ posterior sampling iterations with a burn-in of $b = 125$. Prior parameters were set as follows across both models:

$$\mu_0 = (0, 0, 0, 0)^\top, \qquad \Sigma_0 = 10\, I_4, \qquad s = 2, \qquad \tau = 1,$$
$$\mu_\beta = (0, 0, 0, 0)^\top, \qquad \Sigma_\beta = 10\, I_4.$$

The significance level was set to $\alpha = 0.05$. To compute the surrogate validation threshold η, we set the desired power to $1 - \beta = 0.8$ and used a uniform prior for V_S by setting $a = b = 1$.

References

Schwartz, L. (1965). On bayes procedures. *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete* **4,** 10–26.